

## QUILPIE, Australia

WMO: 944940

Lat: 26.613S

Lon: 144.258E

Elev: 197

StdP: 98.98

Time Zone: 10.00 (E10)

Period: 94-05

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 4.2        | 5.5 | -10.7                         | 1.5 | 24.2 | -8.5 | 1.9 | 21.4 | 8.8                   | 19.1 | 7.9 | 19.3 | 0.7                   | 180  | 0.360 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 11.9                   | 40.6            | 20.2 | 39.3 | 20.1 | 38.0 | 19.9 | 24.8                | 31.0 | 24.0 | 31.0 | 23.2 | 30.4 | 3.1                  | 310  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.2                            | 18.5 | 27.6 | 22.3 | 17.4 | 26.9 | 21.3 | 16.3 | 26.7 | 76.7          | 30.5 | 73.3 | 30.9 | 70.2 | 30.4 | 30.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.9           | 7.7  | 6.3 | 1.8                        | 43.2 | 0.8                | 1.3 | 1.2                                                | 44.1 | 0.8        | 44.9 | 0.3        | 45.6 | -0.3       | 46.5 |    |    |
| (5)               |      |     | -0.5                       | 27.2 | 1.3                | 1.4 | -1.4                                               | 28.3 | -2.2       | 29.1 | -2.9       | 29.9 | -3.8       | 30.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 22.7   | 30.7  | 29.6  | 27.7  | 22.8  | 17.9  | 14.7  | 13.6  | 15.6  | 20.2  | 23.7  | 26.7  | 29.5  |  |  |
|                                                                        | DBStd       | 6.77   | 3.13  | 3.08  | 2.81  | 3.21  | 3.06  | 3.14  | 2.69  | 3.36  | 3.87  | 3.25  | 3.79  | 3.24  |  |  |
|                                                                        | HDD10.0     | 3      | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 1     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | HDD18.3     | 437    | 0     | 0     | 0     | 4     | 46    | 115   | 148   | 99    | 24    | 2     | 0     | 0     |  |  |
|                                                                        | CDD10.0     | 4638   | 641   | 549   | 550   | 384   | 245   | 143   | 114   | 175   | 306   | 426   | 501   | 604   |  |  |
|                                                                        | CDD18.3     | 2031   | 382   | 316   | 292   | 139   | 32    | 7     | 3     | 15    | 80    | 169   | 251   | 346   |  |  |
|                                                                        | CDH23.3     | 26743  | 5644  | 4376  | 3747  | 1368  | 218   | 50    | 24    | 181   | 959   | 1990  | 3197  | 4988  |  |  |
| (13)                                                                   | CDH26.7     | 14799  | 3523  | 2544  | 1993  | 549   | 32    | 2     | 1     | 36    | 382   | 939   | 1769  | 3029  |  |  |
| (14)                                                                   | Wind        | WSAvg  | 2.0   | 2.1   | 2.1   | 1.8   | 1.6   | 1.4   | 1.6   | 2.1   | 2.2   | 2.5   | 2.4   | 2.3   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 365    | 62    | 51    | 36    | 29    | 33    | 17    | 17    | 15    | 13    | 19    | 33    | 35    |  |  |
|                                                                        | PrecMax     | 673    | 335   | 236   | 180   | 318   | 173   | 79    | 96    | 49    | 51    | 55    | 168   | 275   |  |  |
|                                                                        | PrecMin     | 97     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 153    | 66    | 60    | 38    | 58    | 43    | 19    | 21    | 14    | 14    | 17    | 39    | 47    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 42.6  | 40.8  | 39.2  | 35.1  | 29.4  | 26.7  | 25.9  | 30.1  | 35.3  | 38.0  | 41.3  | 41.3  |  |  |
|                                                                        |             | MCWB   | 20.9  | 21.6  | 19.6  | 19.4  | 16.5  | 15.9  | 14.0  | 15.5  | 16.6  | 17.3  | 18.8  | 20.2  |  |  |
|                                                                        | 2%          | DB     | 40.9  | 39.3  | 37.1  | 33.1  | 27.8  | 25.0  | 23.7  | 27.5  | 33.0  | 35.9  | 38.8  | 39.7  |  |  |
|                                                                        |             | MCWB   | 20.5  | 21.9  | 19.5  | 17.9  | 16.1  | 15.6  | 13.4  | 14.5  | 15.9  | 16.8  | 18.3  | 19.5  |  |  |
|                                                                        | 5%          | DB     | 39.4  | 37.7  | 35.6  | 31.6  | 26.0  | 23.2  | 22.1  | 25.7  | 31.3  | 33.9  | 36.7  | 38.5  |  |  |
|                                                                        |             | MCWB   | 20.5  | 21.4  | 19.5  | 17.2  | 15.2  | 14.8  | 12.7  | 13.7  | 15.5  | 16.5  | 18.1  | 19.5  |  |  |
|                                                                        | 10%         | DB     | 37.7  | 36.2  | 34.2  | 30.0  | 24.2  | 21.2  | 20.3  | 23.5  | 29.3  | 31.9  | 34.8  | 37.2  |  |  |
|                                                                        |             | MCWB   | 20.3  | 21.0  | 19.4  | 16.7  | 14.7  | 13.7  | 11.7  | 12.8  | 15.0  | 16.0  | 18.0  | 19.2  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 25.2  | 26.8  | 24.6  | 22.3  | 19.9  | 18.4  | 18.0  | 18.0  | 19.7  | 21.5  | 23.2  | 24.4  |  |  |
|                                                                        |             | MCDB   | 30.9  | 35.1  | 29.3  | 28.8  | 22.0  | 23.3  | 21.4  | 27.4  | 27.7  | 28.4  | 28.7  | 34.8  |  |  |
|                                                                        | 2%          | WB     | 24.2  | 25.4  | 23.4  | 21.2  | 18.5  | 17.2  | 15.4  | 16.0  | 18.2  | 19.7  | 22.2  | 23.1  |  |  |
|                                                                        |             | MCDB   | 31.4  | 32.4  | 29.1  | 27.5  | 22.8  | 21.5  | 20.7  | 23.2  | 25.8  | 27.0  | 27.6  | 32.4  |  |  |
|                                                                        | 5%          | WB     | 23.5  | 24.5  | 22.5  | 19.8  | 17.6  | 16.0  | 14.1  | 14.9  | 17.2  | 18.7  | 21.2  | 22.3  |  |  |
|                                                                        |             | MCDB   | 31.1  | 30.8  | 29.7  | 26.1  | 22.6  | 21.1  | 19.6  | 22.5  | 26.9  | 26.9  | 27.3  | 31.2  |  |  |
|                                                                        | 10%         | WB     | 22.7  | 23.6  | 21.5  | 18.4  | 16.5  | 14.9  | 12.8  | 13.9  | 16.1  | 17.8  | 20.3  | 21.6  |  |  |
|                                                                        |             | MCDB   | 30.5  | 30.2  | 29.8  | 26.4  | 21.8  | 20.4  | 19.0  | 22.1  | 26.9  | 27.6  | 27.6  | 30.9  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.9  | 10.7  | 11.8  | 12.0  | 11.4  | 11.8  | 12.8  | 13.7  | 14.1  | 13.6  | 12.3  | 12.1  |  |  |
|                                                                        |             | MCDBR  | 13.5  | 12.4  | 13.1  | 14.1  | 13.3  | 13.6  | 15.3  | 16.1  | 16.3  | 16.7  | 14.9  | 14.0  |  |  |
|                                                                        | 5% WB       | MCWBR  | 4.3   | 3.6   | 4.3   | 5.1   | 5.3   | 6.3   | 7.3   | 7.3   | 6.0   | 5.8   | 4.6   | 4.7   |  |  |
|                                                                        |             | MCWBR  | 10.6  | 10.0  | 9.9   | 9.5   | 9.0   | 11.0  | 11.6  | 12.6  | 12.3  | 11.8  | 10.3  | 11.3  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.361 | 0.353 | 0.335 | 0.317 | 0.300 | 0.291 | 0.289 | 0.305 | 0.348 | 0.348 | 0.364 | 0.363 |  |  |
|                                                                        | taud        |        | 2.501 | 2.539 | 2.572 | 2.576 | 2.570 | 2.595 | 2.571 | 2.500 | 2.384 | 2.416 | 2.429 | 2.464 |  |  |
|                                                                        | Ebn at Noon |        | 981   | 973   | 962   | 934   | 908   | 897   | 912   | 933   | 933   | 967   | 972   | 981   |  |  |
|                                                                        | Edh at Noon |        | 115   | 108   | 99    | 90    | 82    | 76    | 81    | 95    | 118   | 121   | 123   | 120   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 7.60  | 7.13  | 6.35  | 5.35  | 4.21  | 3.61  | 4.00  | 5.03  | 6.13  | 7.16  | 7.48  | 7.73  |  |  |
|                                                                        | RadStd      |        | 0.41  | 0.59  | 0.37  | 0.30  | 0.19  | 0.22  | 0.26  | 0.22  | 0.42  | 0.44  | 0.57  | 0.47  |  |  |

## Historical Trends

|      |                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only          | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page